

surveillance system; all I am allowed to say is that I felt transported into a James Bond movie—I wondered if the casino was an imitation of the movies or if it was the other way around. Yet, in spite of such sophistication, their risks had nothing to do with what can be anticipated knowing that the business is a casino. For it turned out that the four largest losses incurred or narrowly avoided by the casino fell completely outside their sophisticated models.

First, they lost around \$100 million when an irreplaceable performer in their main show was maimed by a tiger (the show, *Siegfried and Roy*, had been a major Las Vegas attraction). The tiger had been reared by the performer and even slept in his bedroom; until then, nobody suspected that the powerful animal would turn against its master. In scenario analyses, the casino had even conceived of the animal jumping into the crowd, but nobody came near to the idea of insuring against what happened.

Second, a disgruntled contractor was hurt during the construction of a hotel annex. He was so offended by the settlement offered him that he made an attempt to dynamite the casino. His plan was to put explosives around the pillars in the basement. The attempt was, of course, thwarted (otherwise, to use the arguments in [Chapter 8](#), we would not have been there), but I shivered at the thought of possibly sitting above a pile of dynamite.

Third, casinos must file a special form with the Internal Revenue Service documenting a gambler's profit if it exceeds a given amount. The employee who was supposed to mail the forms hid them, instead, for completely unexplainable reasons, in boxes under his desk. This went on for years without anyone noticing that something was wrong. The employee's refraining from sending the documents was truly impossible to predict. Tax violations (and negligence) being serious offences, the casino faced the near loss of a gambling license or the onerous financial costs of a suspension. Clearly they ended up paying a monstrous fine (an undisclosed amount), which was the luckiest way out of the problem.

Fourth, there was a spate of other dangerous scenes, such as the kidnapping of the casino owner's daughter, which caused him, in order to secure cash for the ransom, to violate gambling laws by dipping into the casino coffers.

Conclusion: A back-of-the-envelope calculation shows that the dollar value of these Black Swans, the off-model hits and potential hits I've just outlined, swamp the on-model risks by a factor of close to 1,000 to 1. The casino spent hundreds of millions of dollars on gambling theory and high-tech surveillance while the bulk of their risks came from outside their models.

All this, and yet the rest of the world still learns about uncertainty and probability from gambling examples.